From the Poisson–Bessel Lattice Energy to Reduced Dynamics: PART I–III Foundations and the Two Missing Bridges

Research companion to the Revision 11 working manuscript

20 October 2026

Abstract

We reconstruct the ideal lattice energy without discarding its nonzero Poisson–Bessel modes, then distinguish it from the finite-temperature potential of mean force and from a probability-current law. Explicit collective coordinates, mass and measure give the conditional partition integral. Conditional projection gives a generally nonlinear memory equation. An exactly eliminable harmonic bath keeps the full nonlinear Bessel potential while exhibiting Gaussian but recurrent colored forcing. Separate Markov and small-mass assumptions yield a conditional diffusion, including its noise-induced drift for variable mobility. Eight independent mathematical checks support the identities and counterexamples in declared dimensionless models. They do not calibrate aluminium or establish a fatigue mechanism. Revision 11 is retained as a conditional framework; same-coordinate thermodynamic, kinetic and observation gates remain open.

Scope. First delivery: PART I–III only. The full later mechanism, spatial coupling, 55-display downstream audit and experimental fatigue-life validation are deferred. Original Revision 11, production solver, calibration and historical results are preserved. This is a foundations audit, not a substitution of hypothetical numbers for missing material data.

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1 The two questions

The mathematical identity for an ideal rigid lattice is not the disputed step. The two questions are

$$W_\Gamma^{\mathrm{Bessel}} \xrightarrow[\text{same coordinate and measure}]{\text{integrate out atoms}} A(q) \xrightarrow[\text{time scales and preparation}]{\text{reduce actual dynamics}} \boldsymbol{J}. \tag{1}$$

Here $q = (\delta a, s)$, with normal cut opening and tangential registry. An additive shift to a changes no derivative. A is a free energy, not a cut area. Geometry, coherent patch size, temperature, load, initial ensemble and boundaries must be the same through both arrows.

2 PART I: lattice energy, counting and derivatives

2.1 Hamiltonian and the row transform

For finite atom positions R and momenta Π,

$$H = \tfrac{1}{2}\Pi^T m^{-1}\Pi + V(R), \qquad V_{\mathrm{pair}} = \sum_{i<j} 4\epsilon[(\sigma/r_{ij})^{12} - (\sigma/r_{ij})^6]. \tag{2}$$

The pair force is $-\phi'(r)\widehat{r}$ and its radial Hessian is $\phi''(r)\widehat{r}\widehat{r}^T + (\phi'(r)/r)(I - \widehat{r}\widehat{r}^T)$, with

$$\phi' = 4\epsilon[-12\sigma^{12}r^{-13} + 6\sigma^6 r^{-7}], \tag{3}$$

$$\phi'' = 4\epsilon[156\sigma^{12}r^{-14} - 42\sigma^6 r^{-8}]. \tag{4}$$

For $a > 0$, $p > 1/2$, define $x_j = (j + 1/2)b - s$ and

$$S_p(a, s) = \sum_{j\in\mathbb{Z}} (a^2 + x_j^2)^{-p}, \qquad W_{\mathrm{row}} = 4\epsilon(\sigma^{12}S_6 - \sigma^6 S_3). \tag{5}$$

Using $z^{-p} = \Gamma(p)^{-1}\int_0^\infty t^{p-1}e^{-tz}\mathrm{d}t$, the Gaussian Fourier integral gives, for $k \neq 0$,

$$\widehat{f}_p(k) = \frac{\sqrt{\pi}}{\Gamma(p)}\int_0^\infty t^{p-3/2}e^{-a^2t-k^2/(4t)}\mathrm{d}t \tag{6}$$

$$= \frac{2\sqrt{\pi}}{\Gamma(p)}\left(\frac{|k|}{2a}\right)^{p-1/2} K_{p-1/2}(a|k|), \tag{7}$$

$$\widehat{f}_p(0) = \frac{\sqrt{\pi}\Gamma(p-1/2)}{\Gamma(p)}a^{1-2p}. \tag{8}$$

The last integral is the modified-Bessel representation[1]. Poisson summation then gives

$$S_p = \frac{\sqrt{\pi}\Gamma(p-1/2)}{b\Gamma(p)}a^{1-2p} + \frac{4\sqrt{\pi}}{b\Gamma(p)}\sum_{n=1}^\infty \left(\frac{\pi n}{ab}\right)^{p-1/2} K_{p-1/2}(2\pi na/b)\cos[2\pi n(1/2 - s/b)]. \tag{9}$$

Thus LJ rows use $K_{5/2}$ and $K_{11/2}$. The zero reciprocal mode is a lateral average, not a bulk environment. All nonzero modes are retained. On compact sets bounded away from $a = 0$, the sums and their first two derivative sums converge uniformly. Direct differentiation gives

$$S_{p,a} = -2paS_{p+1}, \qquad S_{p,s} = 2p\sum_j x_j(a^2 + x_j^2)^{-p-1}, \tag{10}$$

$$\begin{aligned} S_{p,aa} &= -2pS_{p+1} + 4p(p+1)a^2S_{p+2}, \\ S_{p,as} &= -4p(p+1)a\sum_j x_j(a^2 + x_j^2)^{-p-2}, \\ S_{p,ss} &= -2pS_{p+1} + 4p(p+1)\sum_j x_j^2(a^2 + x_j^2)^{-p-2}. \end{aligned} \tag{11}$$

2.2 An explicit cut geometry and its multiplicities

Choose the illustrative zigzag stack $r_{j\ell} = (jb + \ell b/2, \ell h)$. Lower layers have $\ell \le 0$; upper layers $\ell \ge 1$ translate by $(-s, \delta a)$. With layer difference $k = \ell_+ - \ell_-$, the correct row-kernel argument is

$$w_k(\delta a, s) = W_{\mathrm{row}}(kh + \delta a, s + (1-k)b/2). \tag{12}$$

There are exactly k crossing layer pairs for each $k \ge 1$, since $1 - k \le \ell_- \le 0$. Hence, per lateral primitive cell,

$$W_\Gamma(\delta a, s) = \sum_{k\ge1} k[w_k(\delta a, s) - w_k(0,0)]. \tag{13}$$

For L lower and U upper layers the count is instead

$$c_k = \max\{0, \min(0, U-k) - \max(1-L, 1-k) + 1\}. \tag{14}$$

No further pair factor 2 or 1/2 applies. Crucially, $\partial_{\delta a}(kh + \delta a) = 1$, so

$$\partial_{\delta a}W_\Gamma = \sum_k k\,\partial_d w_k, \qquad \partial_{\delta a}^2 W_\Gamma = \sum_k k\,\partial_d^2 w_k. \tag{15}$$

The extra chain factor k for a homogeneous dilation $d_k = ka$ is absent. The absolute row baseline converges for $p > 3/2$, including LJ. On $\delta a > -h$, local uniform derivative convergence holds. For opening tending to infinity at fixed registry, dominated convergence gives

$$W_{\mathrm{sep}} = -\sum_{k\ge1} k\,w_k(0,0). \tag{16}$$

The zero mode, all registries and the complete crossing sum enter this limit. Matched equivalent surfaces give $W_{\mathrm{sep}} = 2A_\Gamma\gamma$; rigid cleavage and relaxed surface formation are different targets.

2.3 A full plane is a different transform

For a planar Bravais lattice with primitive area $A_\parallel$,

$$S_p^{(2)}(d, \Delta) = A_\parallel^{-1}\sum_{G\in\Lambda^*} \widehat{f}_p^{(2)}(G)e^{iG\cdot\Delta}, \tag{17}$$

$$\widehat{f}_p^{(2)}(G) = \frac{2\pi}{\Gamma(p)}(|G|/(2d))^{p-1}K_{p-1}(d|G|), \quad G \ne 0, \tag{18}$$

$$\widehat{f}_p^{(2)}(0) = \frac{\pi}{p-1}d^{2-2p}. \tag{19}$$

Here $d > 0, p > 1$. LJ planes use K_2, K_5. The counted plane baseline converges for $p > 2$. For a declared FCC(111) example one can use triangular vectors $a_1 = b(1, 0)$, $a_2 = b(1/2, \sqrt{3}/2)$, $b = a_{\mathrm{lat}}/\sqrt{2}$, $h = a_{\mathrm{lat}}/\sqrt{3}$ and $\tau = (a_1 + a_2)/3$, evaluating the plane kernel at $(kh + \delta a, k\tau + u)$ with count k. Orientation, scalar slip path and translation period remain required inputs. A row certificate is not automatically a full-plane or counted-cut certificate.

For an embedding potential, first form each complete site density and then sum $\sum_i [F(\rho_i(q)) - F(\rho_i^{\mathrm{ref}})]$. Multiplying an individual layer's embedding function by a pair-crossing count is not valid.

3 PART II: atoms, PMF and exact memory

3.1 Collective coordinate, mass and hidden variables

Take a finite, normalizable canonical reference, removing translation zero modes consistently. Define $q = CR + q_c$ with rank-two constant C and

$$G = Cm^{-1}C^T, \qquad M_q = G^{-1}, \qquad B = m^{-1}C^T M_q. \tag{20}$$

Choose $CE = 0$ so $R = R_0 + Bq + Ey$ and $B^T mE = 0$. Then the canonical momentum $p = B^T \Pi = M_q \dot{q}$ separates a kinetic energy $p^T M_q^{-1} p/2$. For normal/tangent projections of two finite group mass-centre differences, $M_q = \mu I_2$, $\mu = M_+ M_-/(M_+ + M_-)$. Other observables have other masses. The hidden variables include internal vibrations, common motion, unretained relative motion and all allowed rearrangements. They are not automatically fast. Infinite half-crystal masses diverge; a finite coherent patch has energy $N_c W_\Gamma$ and mass proportional to N_c. A per-cell potential cannot be combined with an unspecified whole-half-crystal mass.

3.2 The free-energy bridge

Relative to flat $\mathrm{d}q$, define

$$Z(q) = \int \delta(\Xi(R) - q) e^{-\beta V(R)} \mathrm{d}R, \qquad A = -k_{\mathrm{B}} T \log Z + C_A. \tag{21}$$

The affine map has constant Jacobian. With $V(q, y) = U_{\mathrm{Bessel}}(q) + V_\perp(q, y)$, $Z_\perp = \int e^{-\beta V_\perp} \mathrm{d}y$, one obtains exactly

$$\boxed{\Delta A = \Delta U_{\mathrm{Bessel}} - k_{\mathrm{B}} T \log[Z_\perp(q)/Z_\perp(q_0)].} \tag{22}$$

Thus $A = U_{\mathrm{Bessel}} + C_A$ if and only if $Z_\perp$ is constant on the adopted domain and measure. The Bessel identity proves no such condition. The reference units inside logarithms are fixed; differences use ratios.

For nonlinear Ξ, coarea gives

$$Z(q) = \int_{\Xi^{-1}(q)} \frac{e^{-\beta V}}{J_\Xi} \mathrm{d}\sigma_q, \quad J_\Xi = \sqrt{\det(D\Xi D\Xi^T)}. \tag{23}$$

If $b_i \cdot \nabla \Xi_j = \delta_{ij}$, integration by parts gives

$$\partial_i A = \langle b_i \cdot \nabla V - k_{\mathrm{B}} T \nabla \cdot b_i \rangle_q, \tag{24}$$

provided boundary terms vanish. The coordinate measure and constraint ensemble determine metric corrections; do not append a generic correction without those choices[4]. A variable mass metric

along one level set can give a mixture of conditional velocity Gaussians, rather than a single exact inertial mass.

At a unique interior nondegenerate constrained minimum $y_*(q)$, use the measure $w(q,y)\mathrm{d}y$ and positive Hessian $H_\perp$. Laplace expansion gives

$$\begin{aligned} A(q) = U_{\mathrm{rel}}(q) + \tfrac{1}{2}k_{\mathrm{B}}T\log\det H_\perp(q) - k_{\mathrm{B}}T\log w(q,y_*) \\ - \tfrac{d}{2}k_{\mathrm{B}}T\log(2\pi k_{\mathrm{B}}T) + C_A + O(T^2). \end{aligned} \tag{25}$$

Ratios fix units in differences. Zero modes, multiple minima, boundary minima and nonuniform limits need separate treatment. For the exact counterexample $V = U_{\mathrm{Bessel}}(q) + \kappa(q)y^2/2$, $\kappa > 0$,

$$U_{\mathrm{rel}} = U_{\mathrm{rigid}}, \qquad \Delta A = \Delta U_{\mathrm{Bessel}} + \tfrac{1}{2}k_{\mathrm{B}}T\log[\kappa(q)/\kappa(q_0)]. \tag{26}$$

The nonlinear Bessel potential has not been replaced by a parabola; a hidden harmonic degree of freedom changes its PMF even though it leaves the constrained minimum energy unchanged.

3.3 Liouville marginal and conditional projection

For any initial ensemble, the exact marginal continuity law is

$$\partial_t P = -\nabla_q \cdot J, \qquad J = P\mathbb{E}_{\rho_t}[\dot{q} \mid q]. \tag{27}$$

The evolving conditional hidden distribution prevents closure by this identity alone. Smooth finite Hamiltonian positions have zero quadratic variation; Brownian positions require a limiting description.

For an autonomous invariant canonical reference let $Z = (q,p)$, $Lf = \{f, H\}$, $\mathcal{P}f = \mathbb{E}_{\mathrm{eq}}[f \mid Z]$, $\mathcal{Q} = I - \mathcal{P}$. On a suitable common operator domain the Dyson identity gives

$$\dot{Z}(t) = e^{tL}\mathcal{P}LZ + e^{t\mathcal{Q}L}\mathcal{Q}LZ + \int_0^t e^{(t-u)L}\mathcal{P}Le^{u\mathcal{Q}L}\mathcal{Q}LZ\,\mathrm{d}u. \tag{28}$$

For the affine coordinate, $\mathcal{Q}Lq = 0$, $\mathcal{P}Lp = -\nabla A$. Put $F(u) = e^{u\mathcal{Q}L}\mathcal{Q}Lp$ and $C_{ij}(z,u) = \mathbb{E}[F_i(u)F_j(0) \mid Z = z]$. Orthogonality gives $\mathcal{P}F = 0$, not white or independent force. Anti-self-adjointness of L and integration by parts in p give

$$\boxed{\mathcal{P}LF_i(u) = \rho_Z^{-1}\partial_{p_j}(\rho_Z C_{ij}) = \partial_{p_j}C_{ij} - \beta C_{ij}(M_q^{-1}p)_j.} \tag{29}$$

For example, pairing with $g(Z)$ gives $\langle g, LF_i\rangle = -\int \rho_Z \partial_{p_j} g\, C_{ij}\mathrm{d}z$; conditional orthogonality cancels the resolved part of Lg. The memory in (28) evaluates this state-dependent expression at $Z(t-u)$. A velocity convolution follows only in special cases, e.g. C independent of z. Conditional Zwanzig and linear Mori projections must not be mixed without justification[2, 3, 6]. No operator-domain theorem for every infinite LJ lattice is claimed here.

3.4 Exactly eliminable bath atop the full Bessel energy

Consider the specified finite Hamiltonian

$$H = \tfrac{1}{2}p^T M_q^{-1}p + U(q) + \sum_j \left[\frac{P_j^2}{2m_j} + \frac{m_j\omega_j^2}{2}\left(x_j - \frac{c_j^T q}{m_j\omega_j^2}\right)^2\right]. \tag{30}$$

Here U is the full nonlinear Bessel potential. The shifted Gaussian bath integral is independent of q, hence $A = U + C_A$ exactly. Solving each oscillator and integrating by parts gives

$$M_q\ddot{q} = -\nabla U - \int_0^t \Gamma(t-u)\dot{q}(u)\mathrm{d}u + \eta(t) + F_{\mathrm{ext}}(t), \tag{31}$$

$$\Gamma(t) = \sum_j \frac{c_j c_j^T}{m_j\omega_j^2}\cos\omega_j t, \tag{32}$$

$$\eta(t) = \sum_j c_j\left[(x_j(0) - c_j^T q(0)/(m_j\omega_j^2))\cos\omega_j t + \frac{P_j(0)}{m_j\omega_j}\sin\omega_j t\right]. \tag{33}$$

Shifted conditional canonical preparation gives

$$\mathbb{E}\eta = 0, \qquad \mathbb{E}[\eta(t)\eta(u)^T] = k_{\mathrm{B}}T\Gamma(t-u). \tag{34}$$

Finitely many modes yield Gaussian but colored, recurrent forcing. Unshifted preparation instead gives an initial slip $-\Gamma(t)q(0)$. Changing bath parameters changes kinetics while leaving the same A. Off-diagonal memory requires symmetry or evidence to vanish. Arbitrary cyclic forcing of q leaves this special linear bath identity intact; a driven anharmonic solid can instead require a nonstationary, two-time memory and a preparation-dependent fluctuation relation.

4 PART III: conditional stochastic limits and current

4.1 White-bath limit and Kramers dynamics

A chosen exponentially decorrelating family has

$$\Gamma_\varepsilon(t) = \zeta\varepsilon^{-1}e^{-t/\varepsilon},\ t \geq 0, \quad \int_0^\infty \Gamma_\varepsilon \mathrm{d}t = \zeta. \tag{35}$$

Its stationary noise covariance $k_{\mathrm{B}}T\zeta\varepsilon^{-1}e^{-|t-u|/\varepsilon}$ tends to $2k_{\mathrm{B}}T\zeta\delta(t-u)$. This is a controlled bath family, not a spectral derivation for the actual lattice. General non-Gaussian atomic baths need mixing and a functional limit plus feedback/preparation control. With constant symmetric $\zeta > 0$, constant T and constant mass, the candidate Markov inertial equation is

$$\mathrm{d}q = M_q^{-1}p\mathrm{d}t, \tag{36}$$

$$\mathrm{d}p = [-\nabla A + F_{\mathrm{ext}} - \zeta M_q^{-1}p]\mathrm{d}t + \sqrt{2k_{\mathrm{B}}T\zeta}\mathrm{d}B_t. \tag{37}$$

Its Kramers equation retains momentum transport and momentum diffusion. Integrating it over p gives a current moment, not yet a position-only Fick law. The conditions are separately

$$\tau_{\mathrm{bath}} \ll \tau_q, \tau_{\mathrm{load}}, \qquad \tau_p \ll \tau_q, \tau_{\mathrm{load}}, \quad \tau_p = \mu/\zeta \quad \text{(scalar case)}. \tag{38}$$

The first supports Markov approximation; the second supports overdamping. Joint colored-noise and small-mass limits require their ratio/order to be declared. Absorbing boundaries can create additional inertial layers.

4.2 Variable mobility and the forward equation

For constant temperature and a smooth symmetric positive mobility $M(q)$, the controlled small-mass limit of an inertial SDE with $\sigma\sigma^T = 2k_{\mathrm{B}}T\zeta$ and $M = \zeta^{-1}$ is, in Itô form,

$$\boxed{\mathrm{d}q = [-M\nabla A + MF_{\mathrm{ext}} + k_{\mathrm{B}}T\nabla\cdot M]\mathrm{d}t + B\mathrm{d}B_t, \qquad BB^T = 2k_{\mathrm{B}}TM.} \tag{39}$$

Regularity, positive friction, growth/nonexplosion or localization, initial data and boundary assumptions remain required[5]. This theorem starts from an inertial SDE, not from unspecified atomic noise. The Itô forward equation gives

$$J_i = b_iP - k_{\mathrm{B}}T\partial_j(M_{ij}P), \qquad \boxed{J = -M[(\nabla A - F_{\mathrm{ext}})P + k_{\mathrm{B}}T\nabla P].} \tag{40}$$

Omitting the mobility divergence in (39) changes this current. A Stratonovich representation subtracts $\frac{1}{2}\sum_{jk} B_{jk}\partial_j B_{ik}$ from the Itô drift; in multiple dimensions that correction depends on the noise frame and is not universally half the mobility divergence. Constant M, T recover Rev.11's algebra only after all earlier identification assumptions hold.

At fixed conservative external potential and admissible no-flux/decay boundaries, a normalizable Gibbs law is stationary. Integration by parts gives a nonpositive free-energy derivative $-\int P[\nabla(A + V_{\mathrm{ext}}) + k_{\mathrm{B}}T\nabla\log P]^T M[\nabla(A + V_{\mathrm{ext}}) + k_{\mathrm{B}}T\nabla\log P]\mathrm{d}q$. Absorbing populations and time-dependent work require their own extra terms.

4.3 Domain and physical normalization

The ideal rigid cut tends to finite energy at infinite opening and is periodic in unwrapped slip. Its global Gibbs integral on that domain diverges even without tensile load. Equilibrium projection needs a declared finite confining or restricted reference; a periodic quotient loses winding history unless an index is retained. A numerical reflecting box does not prove physical confinement or a global continuous-law contract.

For pure normal traction, $V_{\mathrm{ext}} = -A_\Gamma T_n(t)\delta a$, so $F_s = 0$. The old projected work $-f[(a - a_0) + \chi s]$ has a tangential component χf and requires its own loading geometry. Static energy does not identify a mobility. With $q = \ell q^*$, $A = E_0A^*$, $M = M_0M^*$,

$$t_0 = \ell^2/(M_0E_0), \qquad \theta = k_{\mathrm{B}}T/E_0. \tag{41}$$

A fully specified physical Hamiltonian has Newtonian time; the missing quantity from an energy-only overdamped model is its compatible bath and mobility. Dividing a coherent plane energy by N_p requires both $M \mapsto N_pM$ and $k_{\mathrm{B}}T \mapsto k_{\mathrm{B}}T/N_p$ to preserve the generator. Multiplying mobility alone is not equivalent.

Existing repository v31 studies include actual Al99 MD and finite-band plane-coordinate kinetic estimates. Their error sets include zero drag, and their plane PMF/local-coordinate and thermal normalization match is unresolved. They are evidence, but do not presently calibrate this local physical clock. No measured seconds, hertz or fatigue cycles are inferred.

5 Executed mathematical checks

All parameters below are illustrative and dimensionless. The independent script saved all truncations, tolerances and channel errors; it ran to exit code zero with eight passes and no failures. No production regression or application suite is claimed.

Check	Evidence and scope
Row direct versus Poisson	Maximum scaled value/gradient/Hessian error $9.84\,10^{-15}$; near-singular large absolute values are not assigned that absolute error.
Triangular plane transform	Real/reciprocal cutoffs refined; recorded analytic tail bound covers energy, not a Hessian tail certificate.
Finite crossing counts	4×5 layers, 20 crossing row pairs; direct/counted difference $9.44\,10^{-16}$.
Cut derivatives and endpoint	Stress-free stack $h = 0.761436142462363$; positive Hessian eigenvalues $60.8604, 141.4438$; $W_{\mathrm{sep}} = 2.563620429191983$.
Exact PMF counterexample	At $(0.1, 0.17)$, $k_{\mathrm{B}}T = 0.2$, $\Delta U = 0.5170967738516374$, $\Delta A = 0.5458266268875693$; same Bessel potential, hidden stiffness entropy.
Finite Hamiltonian bath	Refined nonlinear force identity residual $1.23\,10^{-15}$; external-work balance $1.04\,10^{-14}$; analytic covariance, recurrence and initial slip checked.
Two limit controls	Explicit local Bessel-Hessian OU model: equilibrium covariance residual $8.00\,10^{-15}$; separate memory and mass refinements reduce frequency-response errors.
Variable-mobility drift	On a fixed-opening periodic quotient, Gibbs flux residual $3.56\,10^{-15}$ with correct drift; 2.4654 without it.

Two truncations agreeing at machine precision do not prove zero omitted tail. The local quadratic limit control is explicitly an approximation; the finite-bath force check keeps the nonlinear Bessel energy. No Monte Carlo fatigue count replaces a derivation.

6 Revision 11 decision and next gates

The actual Papeeria Rev.11 already calls its diffusion an assumed closure. Its constant-mobility FPE and interior Itô SDE are algebraically consistent. The first eleven displays were audited: row energy, bulk/cut counts, fixed-geometry derivatives, affine work and density transformations can be retained under their stated conditions. The missing links are the thermodynamic identification and the material dynamical limits. The abstract's wording is qualified; the original equations are preserved.

Retain Rev.11 as a conditional reference. First bind the same microscopic coordinate, patch, measure and preparation. Then compare constrained PMF and mean force with the rigid/relaxed energy over the occupied and escape region. Next test conditional memory, inertial response, cross mobility, preparation dependence and loading time scales on held-out trajectories. Only after those gates establish compatible kinetics and boundaries can one assign a physical clock and test an overdamped first-opening law. An absorbing event's increasing CDF does not prove prior microstructural damage accumulation or its equivalence to a detected crack. Later defect, spatial and observation studies remain separate.

A Source and reproducibility record

Repository: `https://github.com/donghyeok8649daniel/al-fatigue-probability-theory`. Base: `c8ccff7553cbb2e66b0c37a76ee8a24808478824`. Files under `docs/research/20261020_first_principles_i_iii/` contain the detailed six requested research notes, Korean executive summary, scope and provenance. Isolated code is under `research/first_principles_20261020/`;

the immutable saved result is `results/run_001/verification.json`. Python 3.12.14, NumPy 2.5.3 and SciPy 1.18.1 were used.

The downloaded Rev.11 source SHA-256 (two contiguous lines) is

```
ce46fad9a667ce0c6ef2e000ff9499148
ff7d20d1a56c427516311a9be2199be
```

The exact source and bibliography are preserved separately. This companion does not silently replace production potentials, calibration or historical outputs. The full 55-display audit and actual-material fatigue validation are outside this first delivery.

References

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